BOUNDARY REGULARITY OF INFINITY-HARMONIC FUNCTIONS

Abstract. We study the Dirichlet boundary regularity of viscosity solutions of the unnormalized infinity Laplace equation. A uniform interior $C^{1,a}$ estimate is taken as an established input, following the October 4, 2026 OpenAI preprint. Under this input, we prove that arbitrary $C^{1,\gamma}$ Dirichlet data on a $C^{1,\gamma}$ boundary patch, with $0 < \gamma \leq 1$, give a gradient that extends continuously to the patch. The principal estimate converts a boundary affine error ε and a normalized geometric and trace seminorm η into a gradient error $C(\varepsilon+\eta)^{a/((1+a)(3+\gamma))}$. Its proof uses explicit anisotropic viscosity subsolutions and supersolutions, including a verification on their nonsmooth sets. We also establish a general transfer from uniform boundary affine decay of order $1+\lambda$ to full boundary $C^{1,\beta}$ regularity with $\beta = a\lambda/(1+a+\lambda)$. Positive Hölder exponents follow for locally constant traces and for nonconstant data admitting uniformly transverse calibrated chords. Flat constant-trace patches retain the interior exponent. For unrestricted Dirichlet data the general conclusion proved here is C^1; a uniform power rate of boundary affine approximation is a sufficient additional estimate for the full Hölder problem.

Draft status. This candidate proof was obtained by our team on October 08, 2026, through extensive dialogue-based training with AI. The team is currently working hard to verify all the proofs.

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1. Introduction

Let $\Omega \subset \mathbb{R}^n$, $n \geq 2$, be a domain. We consider continuous viscosity solutions of

$$-\Delta_\infty u := -\langle D^2 u\, Du, Du\rangle = 0 \quad \text{in } \Omega, \qquad u = g \quad \text{on } \partial\Omega. \tag{1.1}$$

The operator in (1.1) is the unnormalized infinity Laplacian. Its degeneracy persists at nonzero gradients: its coefficient matrix in the Hessian variable has rank one. A boundary proof must therefore use more than the regularity mechanisms available for uniformly elliptic equations.

The objective of this paper is to pass from a uniform interior gradient estimate to regularity at a prescribed boundary patch. In particular, we address nonconstant Dirichlet data without subtracting their tangential affine part from the equation. Such a subtraction changes the operator, and the resulting shifted-gradient equation requires separate estimates.

1.1. Research background and status.

The connection between (1.1) and absolutely minimizing Lipschitz extensions originates in Aronsson's work. Jensen established uniqueness for the Dirichlet problem, and Crandall–Evans–Gariepy developed the comparison-with-cones characterization [1, 2]. These principles provide the elementary Lipschitz and comparison tools used below.

In the plane, Savin proved interior C^1 regularity and Evans–Savin established a uniform interior $C^{1,a}$ estimate for a positive exponent [3, 4]. Evans–Smart proved everywhere interior differentiability in every dimension [5]. The more recent preprint of Xu, submitted in July 2026, states the optimal planar inclusion $C^{1,1/3}_{\mathrm{loc}}$ [11]. Its Theorem 1.1 is an interior statement; Remark 1.5(i) distinguishes that inclusion from a uniform endpoint estimate controlled only by the oscillation. We therefore do not replace the uniform interior input below by a uniform endpoint assertion attributed to that preprint.

The classical boundary results also distinguish differentiability from continuity of the gradient. Wang–Yu proved boundary differentiability under C^1 assumptions in every dimension and boundary C^1 regularity for planar C^2 domains with C^2 data [6]. Hong subsequently proved boundary differentiability when the boundary and the trace are merely differentiable at the point under consideration [7]; a statement of this result also appears as Theorem 1.3 of [8]. Hong's counterexample shows that a C^1 planar boundary and smooth data need not give a continuous boundary gradient [9]. The stated counterexample concerns C^1 geometry and does not, by itself, assign a specified positive Hölder exponent to the boundary normal.

For flat patches with zero trace, odd reflection is available. Lundström–Olofsson–Singh discuss this reflection principle and its planar consequences in Section 3.4 of their 2025 preprint [10]. Their flat zero-trace argument does not provide a general curved-boundary estimate for arbitrary data.

The literature review for this manuscript was updated on October 17, 2026. Among the primary sources examined, we did not identify a theorem establishing full boundary gradient Hölder estimates for arbitrary smooth Dirichlet data that would supply the quantitative decay required in Section 8. This is a statement of the scope of the sources reviewed, not a claim that an exhaustive bibliography has been obtained. All results proved here are formulated relative to the following explicit interior input.

1.2. Standing interior input.

Assumption 1.1 (Uniform interior gradient estimate)**.** For the dimension under consideration there are $a \in (0,1)$ and C_H such that every infinity-harmonic function h in a ball $B_r(x_0)$ satisfies

$$\|Dh\|_{L^\infty(B_{r/2}(x_0))} + r^a [Dh]_{C^{0,a}(B_{r/2}(x_0))} \leq \frac{C_H}{r} \operatorname{osc}_{B_r(x_0)} h. \tag{H}$$

We take Assumption 1.1 as an established theorem, as stipulated for the OpenAI preprint [12]. Its proof is not revisited here. Spatial translations, orthogonal changes of variables, isotropic dilations, multiplication of the dependent variable by a constant, and addition of a

constant preserve the homogeneous equation. These are the transformations used in applying (H).

1.3. **Main results.** By a $C^{1,\gamma}$ boundary patch we mean a relatively open portion $\Gamma \subset \partial\Omega$ represented locally by $C^{1,\gamma}$ graphs. The trace $g|_\Gamma$ is $C^{1,\gamma}$ if it has this regularity in the graph coordinates. If the geometric and trace exponents differ, their minimum can be used throughout. Uniform assertions on a smaller patch $\Gamma' \Subset \Gamma$ refer to fixed chart radii and fixed bounds for the corresponding norms.

Theorem 1.2 (General Dirichlet data)**.** *Assume* (H)*. Let* Γ *and* $g|_\Gamma$ *be* $C^{1,\gamma}$*, where* $0 < \gamma \le 1$*, and let* u *solve* (1.1) *continuously up to* Γ*. Then* Du *extends continuously from* Ω *to* Γ*. The extension agrees with the boundary derivative, and hence*

$$u \in C^1_{\mathrm{loc}}(\Omega \cup \Gamma). \tag{1.2}$$

On smaller closed patches, the gradient has a uniform modulus of continuity over classes with bounded solution oscillation, bounded $C^{1,\gamma}$ *geometry and trace norms, and a common chart radius.*

The quantitative ingredient in this result is as follows. After fixing tangent coordinates at the origin, write

$$\Omega \cap B_1 = \{(x', t) \in B_1 : t > \psi(x')\}, \qquad \psi(0) = 0, \quad D\psi(0) = 0. \tag{1.3}$$

Let $G(x') = g(x', \psi(x'))$ be the coordinate trace, with $G(0) = 0$, $DG(0) = p$. Set

$$[D\psi]_{C^{0,\gamma}} + [DG]_{C^{0,\gamma}} \le \eta. \tag{1.4}$$

The graph and datum are understood to be defined on a slightly larger coordinate patch whenever necessary; only their restrictions to the displayed comparison domain enter the estimates.

Theorem 1.3 (Quantitative boundary flatness estimate)**.** *Assume* (H) *and* (1.3)–(1.4)*. Let* $u \in C(\overline{\Omega \cap B_1})$ *be infinity harmonic in* $\Omega \cap B_1$*, with trace* g *on the graph. Suppose* $e = (p, b)$*,* $|e| \le 1$*, and*

$$\|u - e \cdot x\|_{L^\infty(\Omega \cap B_1)} \le \varepsilon. \tag{1.5}$$

There are $c_* > 0$*,* $E_* > 0$*, and* C*, depending only on* n, γ, a, C_H*, such that* $\varepsilon + \eta \le E_*$ *implies*

$$\sup_{\Omega \cap B_{c_*}} |Du - e| \le C(\varepsilon + \eta)^\theta, \qquad \theta = \frac{a}{(1+a)(3+\gamma)}. \tag{1.6}$$

The power in (1.6) measures dependence on the affine error. It is not asserted to be a spatial Hölder exponent for arbitrary data.

Theorem 1.4 (Uniform boundary affine decay)**.** *Assume* (H)*. Let* u *be infinity harmonic near a* C^1 *boundary patch, continuous up to it with trace* g*, and locally Lipschitz for the ambient Euclidean distance on the closed patch. Suppose there are* $0 < \lambda \le 1$*, common constants* $M, r_0 > 0$*, and a vector* p_z *for each boundary point* z*, independent of* r*, such that for every* z *in the patch and* $0 < r < r_0$*,*

$$\sup_{\overline{\Omega} \cap B_r(z)} |u(x) - g(z) - p_z \cdot (x - z)| \le M r^{1+\lambda}. \tag{1.7}$$

Then, on every smaller closed patch,

$$u \in C^{1,\beta}, \qquad \beta = \frac{a\lambda}{1 + a + \lambda}. \tag{1.8}$$

The estimate is a full pairwise gradient Hölder estimate, including pairs of interior points approaching the boundary.

Theorem 1.5 (Constant traces)**.** *Assume* (H)*, and let* u *solve* (1.1) *continuously up to* Γ*. If* $\Gamma \in C^{1,\gamma}$*,* $0 < \gamma \le 1$*, and* $g = c$ *on* Γ*, then*

$$u \in C^{1,\beta} \quad \text{on smaller closed patches}, \qquad \beta = \frac{a\gamma}{1+a+\gamma}. \tag{1.9}$$

For a flat patch with constant trace, the exponent a *itself holds. For example, in normalized flat coordinates,*

$$\|Du\|_{L^\infty(B^+_{1/2})} + [Du]_{C^{0,a}(\overline{B}^+_{1/2})} \le C\,\|u-c\|_{L^\infty(B^+_1)}. \tag{1.10}$$

Finally, a geometric condition on nonconstant data gives the affine decay in Theorem 1.4. Let ν denote the inward normal.

Theorem 1.6 (Transverse calibrated chords)**.** *Assume* (H)*. Let* Ω *be a bounded convex domain,* g *be globally Lipschitz on* $\partial\Omega$*, and* $L = \mathrm{Lip}(g, \partial\Omega) > 0$*. Let* $u \in C(\overline{\Omega})$ *solve* (1.1)*. Suppose that* Γ *and* $g|_\Gamma$ *are* $C^{1,\gamma}$*. Assume that for each* $z \in \Gamma$ *there are* $b_z \in \partial\Omega$ *and* $\sigma_z \in \{-1, 1\}$ *such that*

$$\sigma_z(g(b_z) - g(z)) = L|b_z - z|, \qquad |b_z - z| \ge \ell > 0, \tag{1.11}$$

and, with $e_z = (b_z - z)/|b_z - z|$*,*

$$e_z \cdot \nu(z) \ge \tau > 0. \tag{1.12}$$

Then uniformly near smaller patches,

$$|u(x) - g(z) - \sigma_z L e_z \cdot (x - z)| \le C|x - z|^{1+\gamma}. \tag{1.13}$$

Consequently $u \in C^{1,\beta}$ *there, where* $\beta = a\gamma/(1 + a + \gamma)$*.*

The calibration in (1.11) is an additional hypothesis on the data. Examples in Section 7 show both that it holds for smooth nonaffine traces and that smooth strict convexity of the domain alone does not imply it.

1.4. **Order of the proof.** Section 2 fixes the viscosity convention and proves the Lipschitz, interpolation, and affine-approximation facts needed later. Section 3 constructs the anisotropic barriers and proves Theorem 1.3. Boundary differentiability and this estimate give Theorem 1.2 in Section 4, including the compactness argument for a uniform modulus. Section 5 proves Theorem 1.4. Section 6 treats constant traces by reflection and comparison with full-ball replacements, and Section 7 proves Theorem 1.6. The last section states a precise additional decay estimate that would yield unrestricted boundary Hölder regularity.

2. Preliminaries

2.1. **Viscosity convention and classical inputs.** For $F(p, X) = -\langle Xp, p\rangle$, a continuous function is a viscosity subsolution if $F(D\phi, D^2\phi) \le 0$ whenever a C^2 function ϕ touches it from above, and a supersolution if the reversed inequality holds for tests from below. In particular, the test inequality at a zero gradient is automatic for the unnormalized homogeneous operator.

We use the following classical facts with their stated scopes [1, 2, 7, 8].

(i) Continuous viscosity subsolutions and supersolutions of the homogeneous equation satisfy comparison on bounded domains when their continuous boundary values are ordered.

(ii) For a ball, continuous boundary data admit a unique continuous infinity-harmonic extension. Locally uniform limits of such solutions are solutions on every compact subset of the limiting open domain.

(iii) Affine functions and cones $A + B|x - z|$ are infinity harmonic away from the vertex. Consequently they can be used in bounded comparison domains not containing the vertex in their interior.

(iv) If a domain is locally a graph differentiable at z, and its Dirichlet trace is differentiable in those coordinates at z, then there is a unique vector p_z for which

$$u(x) = g(z) + p_z \cdot (x - z) + o(|x - z|) \quad (x \in \overline{\Omega},\ x \to z). \tag{2.1}$$

This is the local boundary differentiability theorem [8, Theorem 1.3]. Restricting the expansion to the boundary graph gives its tangential projection $D_T g(z)$.

These are the external boundary and comparison results used in the paper. The estimates and regularity consequences specific to the present argument are proved below.

For a subset $E \subset \mathbb{R}^n$, $\mathrm{Lip}(f, E)$ denotes the Lipschitz constant for the Euclidean distance. Hölder seminorms also use the Euclidean distance. By a smaller closed patch we mean a set $\overline{\Omega} \cap \overline{U'}$, where U' is a bounded open set whose closure lies within the open neighborhood where the stated graph and trace hypotheses hold. We always leave a fixed positive margin to the edge of this neighborhood when choosing comparison balls or nearest boundary points.

2.2. **Geometry, slopes, and local Lipschitz bounds.**

Lemma 2.1 (Geometry of graph patches)**.** *On a fixed smaller portion of a $C^{1,\gamma}$ graph patch, the following properties hold with constants controlled by the chart bounds.*

(i) *The inward normal is $C^{0,\gamma}$. Charts based at their own tangent planes have uniform $C^{1,\gamma}$ bounds.*

(ii) *There are $\kappa > 0$ and $r_1 > 0$ such that for every $x \in \overline{\Omega}$ in the smaller patch and $0 < r < r_1$, the set $\Omega \cap B_r(x)$ contains a ball of radius κr.*

(iii) *For an interior point close to the patch, any nearest boundary point z satisfies $x - z = d(x)\nu(z)$, where $d(x) = \mathrm{dist}(x, \partial\Omega)$. If $y \in B_{d(x)/4}(x)$ and w is any nearest boundary point to y, then*

$$\tfrac{3}{4} d(x) \le d(y) \le \tfrac{5}{4} d(x), \qquad |w - z| \le \tfrac{5}{2} d(x). \tag{2.2}$$

Assertions (ii) *and* (iii) *also hold on a C^1 patch, without a specified Hölder modulus for its normal.*

Proof. For a graph $X(\xi) = (\xi, \psi(\xi))$, $\nu = (-D\psi, 1)/(1+|D\psi|^2)^{1/2}$, so the first normal estimate follows by composition. Orthogonal projection of the graph onto a nearby tangent plane is a $C^{1,\gamma}$ local diffeomorphism with uniformly invertible derivative after the chart is made small. The inverse-function formula and the chain rule give the asserted bounds for recentered charts.

For the second assertion, use tangent charts with graph slope at most $1/10$, and choose r_1 so that all the following balls stay within these charts. If $d(x) \ge r/8$, then $B_{r/8}(x) \subset \Omega \cap B_r(x)$. Otherwise a closest boundary point z is within $r/8$ of x. Set $c = z + (r/2)\nu(z)$. In tangent coordinates at z, every point of $B_{r/8}(c)$ has tangential coordinate of norm less than $r/8$ and normal coordinate greater than $3r/8$, whereas the graph height is at most $r/80$. This ball therefore lies in Ω. It also lies in $B_r(x)$, since $r/8 + r/2 + |z - x| < 3r/4$. Thus $\kappa = 1/8$ works. Uniform continuity of the chart derivatives makes the radius threshold uniform, also for a C^1 patch.

At a nearest point, differentiation of squared distance along the C^1 boundary gives orthogonality to the tangent space. The vector points inward on a sufficiently small graph neighborhood. Finally, d is 1-Lipschitz, and $|w - z| \le d(y) + |y - x| + d(x)$, proving (2.2). No unique nearest-point projection is being asserted. □

Lemma 2.2 (Local Lipschitz control)**.** *Assume* (H)*. Suppose u is infinity harmonic and continuous up to a Lipschitz graph patch, and the trace is Lipschitz in graph coordinates. On a smaller patch, u is Lipschitz for the ambient Euclidean distance. Its Lipschitz constant is bounded by a constant times the trace Lipschitz bound plus the solution oscillation divided by the available chart radius. One also has*

$$[Du]_{C^{0,a}(B_{d(x)/8}(x))} \le C L d(x)^{-a} \tag{2.3}$$

when x is sufficiently close to the smaller patch, where L is the resulting local Lipschitz bound.

Proof. The graph-coordinate Lipschitz bound gives a Euclidean Lipschitz bound for the trace on a smaller graph. Choose nested chart neighborhoods so that all comparison domains, nearest boundary points, and line segments below stay in the larger one. Let $\mathcal{O}$ be the solution oscillation there. Fix a boundary point z in the smaller graph and a common comparison radius R. On the boundary of $\Omega \cap B_R(z)$ the cones $g(z) \pm L_0|y-z|$ order u if L_0 dominates the trace Lipschitz constant and $2\mathcal{O}/R$: the trace bound applies on the graph portion, and the oscillation bound applies on the artificial sphere. The cone vertex z is on the boundary, so comparison yields

$$|u(y) - g(z)| \le L_0|y-z| \tag{2.4}$$

on the smaller local domain, uniformly in z.

For an interior point x with nearest boundary point z and distance d, (2.4) gives $\operatorname{osc}_{B_{d/2}(x)} u \le 3L_0 d$, since every point of that ball has distance at most $3d/2$ from z. Applying (H) on $B_{d/2}(x)$ bounds $|Du(x)|$ by $6C_H L_0$. Points a fixed distance from the boundary are controlled directly by (H).

To obtain an ambient Lipschitz estimate, let $h = |x-y|$ for two points in a still smaller patch. If $h < d(x)/2$, the segment from x to y is interior, and integration of the gradient bound applies. If $h \ge d(x)/2$, choose nearest boundary points z, w. Then $d(x) \le 2h$, $d(y) \le 3h$, and $|z-w| \le 6h$. Using (2.4) at the two points and the trace Lipschitz bound gives $|u(x)-u(y)| \le CL_0 h$. Pairs with a fixed positive separation are controlled by the oscillation. Boundary points follow by continuity. Finally apply (H) to an interior ball of radius $d(x)/4$, using its oscillation bound $Ld(x)/2$. Its half-ball is $B_{d(x)/8}(x)$, giving (2.3). □

Lemma 2.3 (Interpolation from an affine error)**.** *Suppose $v \in C^{1,a}(B_r(x))$, ℓ is affine, $\|v-\ell\|_{L^\infty(B_r(x))} \le A$, and $[Dv]_{C^{0,a}(B_r(x))} \le H$. For $0 < h < r$,*

$$|Dv(x) - D\ell| \le \frac{2A}{h} + \frac{H}{1+a}h^a. \tag{2.5}$$

In particular, if $A \le M\delta r$ and $H \le Mr^{-a}$, $0 < \delta \le 1$, then $|Dv(x) - D\ell| \le C_a M \delta^{a/(1+a)}$.

Proof. For every unit vector ζ, compare $(v(x+h\zeta) - v(x))/h$ with $D\ell \cdot \zeta$. Their difference is at most $2A/h$. Integration of Dv along the segment bounds the difference from $Dv(x)\cdot\zeta$ by $Hh^a/(1+a)$. Take the supremum over ζ. Choosing h to be a fixed small multiple of $r\delta^{1/(1+a)}$ gives the final assertion. □

Lemma 2.4 (Recentered affine slopes)**.** *Under (1.3)–(1.4), let $e = (DG(0), b)$ with $|e| \le 1$. At nearby boundary points set*

$$a_z = e + D_T g(z) - P_{T_z\partial\Omega} e = D_T g(z) + (e \cdot \nu(z))\nu(z). \tag{2.6}$$

For η small, on a fixed smaller patch,

$$|a_z - e| \le C\eta, \qquad |a_z - a_w| \le C\eta|z-w|^\gamma, \tag{2.7}$$

$$|g(w) - g(z) - a_z \cdot (w-z)| \le C\eta|w-z|^{1+\gamma}. \tag{2.8}$$

In tangent coordinates at each such z, on a common fixed-radius ball, the graph and trace derivative seminorms are $O(\eta)$. If (1.5) holds, the normalized affine error relative to $g(z) + a_z\cdot(x-z)$ is $O(\varepsilon+\eta)$ on that ball.

Proof. For $X(\xi) = (\xi, \psi(\xi))$, put $J = DX$. The tangential gradient is $D_T g(X(\xi)) = J(J^TJ)^{-1}DG(\xi)$. Writing $V = J(J^TJ)^{-1}$, the tangential projection is VJ^T, so

$$a_{X(\xi)} - e = V(\xi)(DG(\xi) - p - bD\psi(\xi)).$$

The parenthesized term vanishes at zero and has supremum and $C^{0,\gamma}$ seminorm bounded by $C\eta$. The matrices V are uniformly bounded and have $C^{0,\gamma}$ seminorm $O(\eta)$. This proves both estimates in (2.7). For $z = X(\xi)$ and $w = X(\zeta)$, the identity $a_z \cdot J(\xi) = DG(\xi)$ gives

$$\begin{aligned} g(w) - g(z) - a_z\cdot(w-z) = {} & G(\zeta) - G(\xi) - DG(\xi)\cdot(\zeta-\xi) \\ & - a_z \cdot (X(\zeta) - X(\xi) - J(\xi)(\zeta-\xi)). \end{aligned}$$

Taylor's formula, $|a_z| \le 2$, and $|\zeta - \xi| \le |w - z|$ prove (2.8). For the recentered charts, use the tangent-projection construction in Lemma 2.1. Its coordinate maps have uniformly invertible derivatives with Hölder variation $O(\eta)$; the inverse derivative formula preserves this bound. The chain rule then gives the asserted $O(\eta)$ seminorms for both the new graph and its trace. After shrinking the patch, their comparison balls have a common fixed radius and lie within B_1.

Finally, (1.5) also holds at z by continuity, and

$$|u(y) - g(z) - a_z \cdot (y - z)| \le 2\varepsilon + |a_z - e|\,|y - z|.$$

Dividing by the fixed comparison radius changes only the constant. The slope may now have size at most 2 instead of 1; the barrier construction below is made with this larger bound in mind. □

2.3. **Affine approximation on clipped balls.**

Lemma 2.5 (Affine approximation criterion)**.** *Let $E = \overline{\Omega}$ on a graph patch satisfying the thickness property of Lemma 2.1. Suppose that for every center x in a fixed smaller closed patch and every $0 < r < r_1$, there is an affine function $P_{x,r}$ such that $P_{x,r}(x) = u(x)$ and*

$$\sup_{E \cap B_r(x)} |u - P_{x,r}| \le M r^{1+\beta}, \qquad 0 < \beta \le 1. \tag{2.9}$$

Then u has a first-order expansion on the closed patch with gradient $p(x)$. For $|x - y| < r_1/4$ in a further smaller patch, $|p(x) - p(y)| \le CM|x - y|^\beta$. In particular, $[p]_{C^{0,\beta}} \le CM$ on further patches of diameter less than $r_1/4$. At interior points $p = Du$. The constant C may depend on n, β and the thickness constant κ.

Proof. If an affine function A is bounded by N on a ball of radius κr, differences at the ends of its diameters give $|DA| \le N/(\kappa r)$. Apply this to $P_{x,r} - P_{x,r/2}$ on the interior ball contained in $E \cap B_{r/2}(x)$. Equation (2.9) gives

$$|DP_{x,r} - DP_{x,r/2}| \le CMr^\beta.$$

Fix the dyadic radii $r_k = 2^{-k-1} r_1$, $k = 0, 1, \ldots$. The gradients converge along these scales to a vector $p(x)$, since

$$\sum_{j=k}^{\infty} |DP_{x,r_j} - DP_{x,r_{j+1}}| \le \frac{CMr_k^\beta}{1 - 2^{-\beta}}.$$

For any $0 < r < r_1$, choose r_k with $r/2 \le r_k \le r$. On $E \cap B_{r_k}(x)$, the difference $P_{x,r} - P_{x,r_k}$ is bounded by $CMr^{1+\beta}$. The interior ball of radius κr_k there therefore gives $|DP_{x,r} - DP_{x,r_k}| \le CMr^\beta$. Combining this with the dyadic estimate proves $|p(x) - DP_{x,r}| \le C_\beta M r^\beta$ for every such radius. Hence

$$\sup_{E \cap B_r(x)} |u(y) - u(x) - p(x) \cdot (y - x)| \le CMr^{1+\beta}.$$

The vector $p(x)$ is unique: the difference of two first-order expansions is an affine function bounded by $o(r)$ on $E \cap B_r(x)$, and the thickness ball bounds its gradient by $o(r)/(\kappa r) \to 0$.

Let $h = |x - y|$ be small and set $R = 4h$. The clipped ball $E \cap B_h(x)$ lies in both $B_R(x)$ and $B_R(y)$ and contains an interior ball of radius κh. The difference $P_{x,R} - P_{y,R}$ is bounded by $CMR^{1+\beta}$ there. Its gradient is therefore bounded by CMh^β. Combining this with the convergence estimate at the two centers proves $|p(x) - p(y)| \le CMh^\beta$. The claimed identification at interior points follows from the first-order expansion. Continuity of p then identifies its boundary values with the limits of Du from the interior. In applications where $|p| \le L$, pairs separated by at least $r_1/4$ have Hölder quotient at most $2L(4/r_1)^\beta$. Thus the local conclusion gives a full seminorm on any fixed smaller patch, with this additional controlled term. □

3. Anisotropic barriers and quantitative boundary flatness

The construction in this section uses the true graph boundary. It does not replace the equation by one obtained through a nonlinear flattening or affine subtraction.

3.1. **A family of viscosity barriers.**

Lemma 3.1 (Anisotropic subsolutions and supersolutions)**.** *Fix $q \in (1,2]$. Let $0 < \delta < \delta_0$, $v = mw$, $|w| = 1$, and $\delta/2 \le m \le 3$. There are constants $\mu, k > 0$, depending only on q, for which the following construction works. Set*

$$\lambda = \mu^2\delta^2, \qquad K = k\delta\sqrt{\lambda}, \qquad A = \frac{4}{q-1}K\lambda^{q/2}, \tag{3.1}$$

and, writing $z = w \cdot x$, $\xi = P_{w^\perp}x$ and $S = (|\xi|^2 + \lambda z^2)^{1/2}$, define

$$\mathcal{B}_v^{\pm}(x) = mz \pm KS^q \mp A|z|^q. \tag{3.2}$$

In B_1, $\mathcal{B}_v^+$ is a viscosity supersolution and $\mathcal{B}_v^-$ a viscosity subsolution of $-\Delta_\infty u = 0$. Moreover

$$|D\mathcal{B}_v^{\pm} - v| \le C_q\delta^2 \quad \text{on } B_1. \tag{3.3}$$

The constant μ may first be made as small as desired; k is then chosen sufficiently small.

Proof. We take $\mu \le 1$ and $\delta_0 \le 1/4$. The functions are C^1 because $q > 1$. For $F = S^q$ and $Q = \operatorname{diag}(I, \lambda)$,

$$D^2F = qS^{q-2}Q + q(q-2)S^{q-4}(Qx)\otimes(Qx) \le qS^{q-2}Q \tag{3.4}$$

away from the origin. Also $|DF| \le qS^{q-1} \le q$ on B_1. Since $K = O_q(\delta^2)$ and $A = O_q(\delta^{q+2})$, (3.3) follows. Also $\mathcal{B}_v^-(x) = -\mathcal{B}_v^+(-x)$. Reflection of the independent variable preserves the equation, and negation interchanges viscosity subsolutions and supersolutions. It therefore suffices to check the upper barrier.

Consider the upper barrier and set $P = D\mathcal{B}_v^+$. For $z \neq 0$,

$$P_\xi = KqS^{q-2}\xi, \qquad P_z = m + Kq\lambda zS^{q-2} - Aq|z|^{q-2}z.$$

The bound $S \ge \sqrt{\lambda}|z|$ yields

$$|P_z - m| \le q\left(1 + \frac{4}{q-1}\right)K\lambda^{q/2}.$$

Decreasing k ensures $P_z \ge m/2$. From (3.4) we obtain

$$\begin{aligned}\Delta_\infty\mathcal{B}_v^+ &\le q^3K^3S^{3q-4} + (Kq\lambda^{q/2} - Aq(q-1))|z|^{q-2}P_z^2\\ &= q^3K^3S^{3q-4} - 3qK\lambda^{q/2}|z|^{q-2}P_z^2.\end{aligned} \tag{3.5}$$

There is one absorption inequality valid for the entire interval $1 < q \le 2$:

$$\begin{aligned}S^{3q-4}|z|^{2-q}\lambda^{1-q/2} &= S^{3q-4}(\sqrt{\lambda}|z|)^{2-q}\\ &\le S^{2q-2} \le 1.\end{aligned} \tag{3.6}$$

Thus

$$\Delta_\infty\mathcal{B}_v^+ \le qK\lambda^{q/2}|z|^{q-2}\left(q^2K^2/\lambda - \tfrac{3}{4}m^2\right).$$

Since $m \ge \delta/2$, the choice $k \le 1/(8q)$ guarantees $K^2 \le m^2\lambda/(16q^2)$. The last display is strictly negative.

For $q < 2$, consider a point $(\xi_0, 0)$ on the remaining plane. If $\xi_0 \neq 0$, restriction to the line parallel to w gives

$$\mathcal{B}_v^+(\xi_0, h) = \mathcal{B}_v^+(\xi_0, 0) + mh - A|h|^q + O(h^2).$$

A C^2 lower test must have first derivative m on that line, so subtracting it leaves $-A|h|^q + O(h^2) < 0$ for small nonzero h. This contradicts touching from below because $q < 2$. At the origin the corresponding coefficient is $K\lambda^{q/2} - A < 0$, so there is again no lower C^2 test. These are precisely the tests required for the supersolution inequality.

When $q = 2$, the upper barrier is a quadratic polynomial. Its infinity Laplacian is continuous, so the inequality already proved on $\{z \neq 0\}$ extends to $\{z = 0\}$. The identity relating the two barriers then gives the lower-barrier assertion in every case. □

3.2. **Ordering on the entire boundary of the comparison domain.**

Lemma 3.2 (Boundary sandwich)**.** *Let $u \in C(\overline{\Omega \cap B_1})$ be infinity harmonic in $\Omega \cap B_1$ with trace g on the graph. Assume (1.3)–(1.5), allowing $|e| \le 2$, and put $q = 1 + \gamma$. Let $0 < \delta < \delta_0$ be in the range of Lemma 3.1. There is $c_q > 0$ such that, whenever*

$$\varepsilon + \eta \le c_q \delta^{q+2}, \tag{3.7}$$

one can choose a supersolution B_+ and a subsolution B_- from Lemma 3.1 with

$$B_- \le u \le B_+ \quad \text{in } \Omega \cap B_1, \qquad |DB_\pm - e| \le C_q \delta. \tag{3.8}$$

Both barriers vanish at the origin.

Proof. First choose $\mu \le 1$ so small that

$$\frac{4}{q-1}(24\mu)^q \le \frac{1}{4}.$$

Next choose k as in Lemma 3.1, also requiring $\frac{4}{q-1} k \mu^{q+1} \delta_0^q \le 1/24$. These choices are fixed independently of δ; the smallness constant c_q will be chosen last. Choose $s_+, s_- \in \{\delta, 2\delta\}$ so that $|b + s_+| \ge \delta/2$ and $|b - s_-| \ge \delta/2$. Such choices exist because the two candidates in either pair differ by δ and cannot both have absolute value less than $\delta/2$. Define $v_\pm = e \pm s_\pm e_n$. For δ_0 small, $m_\pm = |v_\pm|$ lie between $\delta/2$ and 3. Take $B_+ = \mathcal{B}^+_{v_+}$ and $B_- = \mathcal{B}^-_{v_-}$.

On the true boundary $x = (x', \psi(x'))$, with $r = |x'|$,

$$|\psi(x')| + |G(x') - p \cdot x'| \le C\eta r^q. \tag{3.9}$$

For either $v = mw$,

$$|P_{w^\perp}(x', 0)|^2 = r^2 - (w' \cdot x')^2 \ge w_n^2 r^2.$$

The triangle inequality therefore gives

$$|P_{w^\perp} x| \ge (|v_n|/m) r - |\psi(x')| \ge \frac{\delta}{4m} r.$$

Here (3.7), with a small fixed constant, ensures the final inequality. Since $|x| \le 2r$,

$$KS^q - A|z|^q \ge \left(K(\delta/(4m))^q - 2^q A\right) r^q. \tag{3.10}$$

Our choice of μ ensures

$$\frac{2^q A}{K(\delta/(4m))^q} = \frac{4}{q-1}(8m\mu)^q \le \frac{1}{4} \qquad (m \le 3).$$

The remaining coefficient in (3.10) is therefore at least $\frac{3}{4} k \mu 12^{-q} \delta^{q+2}$. It absorbs the errors (3.9) and the normal terms $(b \pm s_\pm)\psi$ after choosing c_q small, since $|b \pm s_\pm| \le 3$. Therefore $B_- \le G \le B_+$ on the true boundary.

On the spherical boundary $\partial B_1 \cap \overline{\Omega \cap B_1}$, put $t = x_n \ge -C\eta$; the last inequality follows by approaching from the graph domain. For either sign, the relevant gap has the form

$$s\,t + KS^q - A|z|^q. \tag{3.11}$$

Set $\alpha = |v_n|/m \ge \delta/(2m) \ge \delta/6$. If $t \ge \alpha/2$, the tilt is at least $\delta^2/(4m) \ge \delta^2/12$. Since $|z| \le 1$ and $A = \frac{4}{q-1} k \mu^{q+1} \delta^{q+2}$, our choice of k gives $A \le \delta^2/24$ for every $\delta < \delta_0$. The gap is then at least $\delta^2/24$, and hence at least a fixed positive multiple of δ^{q+2}.

In the other case, smallness of η implies $t \ge -C\eta \ge -\alpha/2$, so $|t| \le \alpha/2$. Here it suffices that $C\eta \le \delta/12$, as follows from (3.7) after choosing c_q small. For unit vectors w, x,

$$|w_n| \le |x_n| + |P_{w^\perp} x|.$$

Indeed, decompose w into its component parallel to x and its orthogonal component. Thus $S \ge \alpha/2$. The positive term in (3.11) is at least $K(\delta/(4m))^q$; the negative term $A|z|^q$ is

absorbed by the same choice of μ as on the graph. The possible negative tilt costs at most $C\delta\eta$ and is absorbed by (3.7). Hence

$$B_+ - e\cdot x \geq c_q'\delta^{q+2}, \qquad e\cdot x - B_- \geq c_q'\delta^{q+2}$$

on the spherical portion, where $c_q' > 0$ depends only on the fixed parameters. Choose c_q small enough for all the preceding absorptions and $c_q \leq c_q'/2$. Then (1.5) gives the ordering there too. Comparison proves the sandwich. The gradient assertion follows from (3.3) and $|v_\pm - e| \leq 2\delta$. □

3.3. **Proof of the quantitative flatness theorem.**

Proof of Theorem 1.3. Set $E = \varepsilon + \eta$. The case $E = 0$ is immediate. Choose

$$\delta = C_q E^{1/(3+\gamma)} \tag{3.12}$$

with a fixed sufficiently large constant; reduce E_* so that $\delta < \delta_0$. Lemma 2.2 gives a uniform local Lipschitz bound and (2.3): indeed, the normalized assumptions give $\operatorname{osc}_{\Omega\cap B_1} u \leq 2 + 2\varepsilon$ and $\|DG\|_\infty \leq 1 + \eta$.

At each nearby boundary point z, use the slope a_z from Lemma 2.4. Choose a common fixed comparison radius r_0 and an orthogonal map R_z taking e_n to the inward normal at z. The normalized function

$$U_z(X) = \frac{u(z + r_0 R_z X) - g(z)}{r_0}$$

is infinity harmonic, with affine slope $R_z^T a_z$ of size at most 2. Its affine error is at most $C(\varepsilon/r_0 + \eta) \leq CE$, since r_0 is fixed. The recentered graph and trace derivative seminorms are at most $Cr_0^\gamma\eta \leq C\eta$. Thus Lemma 3.2, with the constants in (3.12) enlarged once, applies at every such z. In original normalized coordinates it gives

$$|u(y) - g(z) - a_z\cdot(y - z)| \leq C\delta|y - z| \tag{3.13}$$

whenever y is in that fixed local ball. This follows by integrating the barrier gradient bounds along the segment from their common contact point z; the barriers themselves are defined on the full comparison ball.

Choose the fixed target radius $c_* > 0$ sufficiently small. For $x \in \Omega\cap B_{c_*}$, put $d = \operatorname{dist}(x, \partial\Omega)$ and choose a nearest boundary point z. Since $0 \in \partial\Omega$, $d \leq |x|$ and $|z| \leq 2|x|$. We may therefore ensure that z lies in the recentered patch and $B_{d/4}(x) \subset B_{r_0}(z)$ by shrinking c_*, independently of E. We also take c_* below the fixed distance threshold in (2.3). Let $A_z(y) = g(z) + a_z\cdot(y - z)$. For $y \in B_{d/4}(x)$, $|y - z| \leq 5d/4$, so (3.13) at this single fixed foot directly gives

$$\|u - A_z\|_{L^\infty(B_{d/4}(x))} \leq C\delta d. \tag{3.14}$$

On $B_{d/8}(x)$, (2.3) and Lemma 2.3, using (3.14) and $h = cd\delta^{1/(1+a)}$ with fixed $0 < c < 1/8$, now give

$$|Du(x) - a_z| \leq C\delta^{a/(1+a)}.$$

Since $\eta \leq E \leq C\delta^{3+\gamma}$ and $\delta < 1$, (2.7) absorbs $a_z - e$ into the same bound. Substitution of (3.12) proves (1.6) throughout $\Omega \cap B_{c_*}$. □

4. Boundary continuity of the gradient

4.1. **Pointwise extension.**

Proof of the local conclusion of Theorem 1.2. Fix a smaller patch. Lemma 2.2 gives a fixed ambient Lipschitz bound L. The classical boundary differentiability theorem (2.1) gives, at every boundary point z, a vector p_z and

$$E_z(r) := \frac{1}{r}\sup_{\overline{\Omega}\cap B_r(z)} |u(x) - g(z) - p_z\cdot(x - z)| \longrightarrow 0. \tag{4.1}$$

These vectors are uniformly bounded on the smaller patch. Indeed, for every unit vector ζ with $\zeta\cdot\nu(z) > 0$, differentiability of the graph gives $z + s\zeta \in \Omega$ for all small $s > 0$. Divide

$|u(z+s\zeta)-g(z)|\le Ls$ by s and use (2.1) to obtain $|p_z\cdot\zeta|\le L$. Continuity gives the same inequality on the closed inward hemisphere; its supremum is $|p_z|$, since absolute values allow either sign of each direction. Thus $|p_z|\le L$.

Choose the fixed amplitude scale $M_0=\max\{1,L\}$. In tangent coordinates at z consider

$$u_{z,r}(y)=\frac{u(z+rRy)-g(z)}{M_0 r},$$

where R is the orthogonal map from the tangent coordinates to the original coordinates, with $Re_n=\nu(z)$. The slope is R^Tp_z/M_0, of size at most 1; its tangential component is the derivative of the normalized trace by (2.1). If ψ_z, G_z denote the unscaled tangent graph and coordinate trace, their normalized versions satisfy

$$[D\psi_{z,r}]_{C^{0,\gamma}}=r^\gamma[D\psi_z]_{C^{0,\gamma}},\qquad [DG_{z,r}]_{C^{0,\gamma}}=\frac{r^\gamma}{M_0}[DG_z]_{C^{0,\gamma}},$$

with the unscaled seminorms restricted to the corresponding radius-r coordinate patches. Their sum is at most Kr^γ, where K depends on the fixed geometry and normalized data. The affine error is $E_z(r)/M_0$. Theorem 1.3 therefore applies for all sufficiently small r. Since $Du_{z,r}=R^TDu(z+rRy)/M_0$, it gives

$$\sup_{\Omega\cap B_{c_*r}(z)}|Du-p_z|\le CM_0\left(\frac{E_z(r)}{M_0}+Kr^\gamma\right)^\theta,\qquad \theta=\frac{a}{(1+a)(3+\gamma)}.\tag{4.2}$$

The right side tends to zero.

It remains to identify the extension as a continuous field also along the boundary. Given $\epsilon>0$, choose r so that the left side of (4.2) is at most ϵ. If $w\in\Gamma\cap B_{c_*r/2}(z)$, approach w through interior points in $B_{c_*r}(z)$. The already proved interior limit at w equals p_w, so $|p_w-p_z|\le\epsilon$. Thus the extended gradient is continuous on the closed patch. □

4.2. **Uniformity over bounded classes.**

Proof of the uniform-modulus assertion in Theorem 1.2. Fix a common chart radius, a positive γ, upper bounds for the $C^{1,\gamma}$ chart and trace norms, and an upper bound for the solution oscillations. All conclusions below are on a fixed smaller patch, leaving a fixed margin to a larger controlled chart. Translations and orthogonal changes of variables place the graph representations on common coordinate balls; these changes preserve gradient differences and Euclidean distances. After subtraction of constants, Lemma 2.2 gives uniformly bounded, uniformly Lipschitz solutions. Divide the entire class by one common amplitude scale $M_0\ge 2\max\{1,L\}$, where L bounds these Lipschitz constants. We keep the notation u_j for the rescaled solutions; their boundary slopes have size at most $1/2$. All estimates can be rescaled back with this fixed factor. On a larger closed chart portion E_j, each rescaled solution has Lipschitz constant at most $1/2$. Hence

$$\mathcal{U}_j(x)=\inf_{y\in E_j}\left(u_j(y)+\tfrac12|x-y|\right)$$

is an ambient $1/2$-Lipschitz extension, agreeing with u_j on E_j. The subtracted constants can be chosen to make these extensions uniformly bounded on the common ball.

If no uniform modulus existed, there would be solutions u_j and points x_j, y_j with $|x_j-y_j|\to0$ but $|Du_j(x_j)-Du_j(y_j)|\ge\epsilon_0>0$. Boundary derivatives here mean the continuous extensions just established. Pass to a subsequence so that x_j, y_j tend to a common point x_*. Compactness of bounded $C^{1,\gamma}$ families in C^1 and Arzelà–Ascoli give, on the larger chart,

$$\psi_j\to\psi_\infty,\quad G_j\to G_\infty\quad\text{in } C^1,$$
$$\mathcal{U}_j\to\mathcal{U}_\infty\quad\text{uniformly.}$$

The chart and trace limits retain the common $C^{1,\gamma}$ bounds. Writing $X_j(\xi)=(\xi,\psi_j(\xi))$, the identities $\mathcal{U}_j(X_j(\xi))=G_j(\xi)$ and the common Lipschitz bound give $\mathcal{U}_\infty(X_\infty(\xi))=G_\infty(\xi)$. The restriction u_∞ of $\mathcal{U}_\infty$ to the limiting closed graph domain therefore has this trace and Lipschitz constant at most $1/2$. It is infinity harmonic on the limiting open graph domain:

each compact interior set lies in all sufficiently large approximating domains, where viscosity stability applies. The graph inequalities at x_j also show that x_* belongs to the limiting closed domain.

If the common limit of x_j, y_j is interior, (H) on a common fixed interior ball gives a uniform Hölder bound for the gradients, contradicting their separation. Otherwise translate the common boundary limit to the origin and set $z_j = (0, \psi_j(0)) \to 0$. Then both $x_j - z_j$ and $y_j - z_j$ tend to zero. The limit has a boundary derivative p_∞ by (2.1); the Lipschitz-to-slope argument above gives $|p_\infty| \le 1/2$. For any prescribed small $t > 0$, choose a fixed small r so that the normalized limiting affine error on B_{2r} and the common rescaled seminorm bound Kr^γ are at most t.

Adjust the slope to $a_j = p_\infty + D_T g_j(z_j) - P_{T_{z_j}\partial\Omega_j} p_\infty$. Its tangential component agrees exactly with that of the jth trace, while its normal component is that of p_∞. The adjustment tends to zero because the charts and traces converge in C^1 and $P_{T_0\partial\Omega_\infty} p_\infty = D_T g_\infty(0)$; in particular $|a_j| \le 1$ for large j.

For every x in the jth closed graph domain in $B_r(z_j)$, the point $x^\infty = (x', \max\{x_n, \psi_\infty(x')\})$ belongs to the limiting closed graph domain, satisfies $|x - x^\infty| \le \|\psi_j - \psi_\infty\|_\infty = o_j(1)$, and lies in B_{2r} for large j. Uniform convergence of the ambient extensions, their Lipschitz bounds, and the limiting affine error give

$$|u_j(x) - g_j(z_j) - p_\infty \cdot (x - z_j)| \le 2tr + o_j(1).$$

Here the uniform $o_j(1)$ includes the moving intercept $g_j(z_j) - g_\infty(0)$, the base point z_j, and the graph and extension discrepancies. Consequently the normalized error for a_j is

$$\begin{aligned}\varepsilon_j &:= \frac{1}{r} \sup_{\overline{\Omega}_j \cap B_r(z_j)} |u_j(x) - g_j(z_j) - a_j \cdot (x - z_j)| \\ &\le 2t + \frac{o_j(1)}{r} + |a_j - p_\infty|.\end{aligned}$$

The radius r is fixed before taking $j \to \infty$. In tangent coordinates at z_j, the graph and trace seminorms are at most $Kr^\gamma \le t$, uniformly in j. Apply the same normalization as above, now with amplitude scale 1. The slope is the tangent-coordinate expression of a_j and its tangential part matches the rescaled trace. For t sufficiently small and then j sufficiently large, Theorem 1.3 gives, in the original coordinates,

$$\sup_{\Omega_j \cap B_{c_* r}(z_j)} |Du_j - a_j| \le Ct^\theta + o_j(1).$$

The estimate extends to boundary gradients by their continuity for each j. Both x_j and y_j are in this ball for large j. Taking t sufficiently small contradicts their separation. This proves a uniform modulus. □

Remark 4.1. The modulus obtained by compactness need not be a power. Formula (4.2) is quantitative in the affine error $E_z(r)$, while (2.1) gives only $E_z(r) \to 0$. Neither step asserts $E_z(r) \le Cr^\lambda$. The constants in the barrier argument depend on a fixed positive γ and can deteriorate as $\gamma \downarrow 0$; the theorem therefore makes no assertion for an arbitrary merely C^1 modulus.

5. From boundary affine decay to gradient Hölder regularity

Proof of Theorem 1.4. Work in normalized local coordinates and on a patch slightly larger than the final one. Let L be a local ambient Lipschitz bound on that larger patch. The C^1 case of Lemma 2.1 supplies the uniform thickness property on the smaller patch. For a target radius ρ, set

$$r = \rho^{(1+a)/(1+a+\lambda)}. \tag{5.1}$$

Since $0 < (1 + a)/(1 + a + \lambda) < 1$, there is a fixed $\rho_* > 0$ such that, for $0 < \rho < \rho_*$, one has $r \ge 8\rho$ and $2r < r_0$. Choose ρ_* also below the thickness radius and so small that all balls

and nearest boundary points below remain in the larger controlled patch. These choices are uniform over the centers.

If an interior center x has $d(x) \geq r$, apply (H) on $B_{r/2}(x)$, using $\operatorname{osc}_{B_{r/2}(x)} u \leq Lr$, to obtain

$$[Du]_{C^{0,a}(B_{r/4}(x))} \leq CLr^{-a}.$$

Integrating this estimate along the segment from x to $y \in B_\rho(x) \subset B_{r/4}(x)$ gives

$$\sup_{B_\rho(x)} |u(y) - u(x) - Du(x) \cdot (y - x)| \leq CL\rho^{1+a} r^{-a}. \tag{5.2}$$

If $d(x) < r$, choose a nearest boundary point z; for a boundary center take $z = x$. Then $\overline{\Omega} \cap B_\rho(x) \subset \overline{\Omega} \cap B_{2r}(z)$. Write $Q_z(y) = g(z) + p_z \cdot (y - z)$ and define

$$P_{x,\rho}(y) = Q_z(y) + u(x) - Q_z(x).$$

Then $P_{x,\rho}(x) = u(x)$. Both x and the clipped target ball lie in $\overline{\Omega} \cap B_{2r}(z)$, so (1.7) gives error at most

$$2M(2r)^{1+\lambda} \leq CMr^{1+\lambda} \tag{5.3}$$

on that clipped ball. The two terms come from the affine errors at y and at the anchoring point x.

For the choice (5.1),

$$\rho^{1+a} r^{-a} = r^{1+\lambda} = \rho^{1+\beta}, \qquad \beta = \frac{a\lambda}{1 + a + \lambda}.$$

Here $0 < \beta < a < 1$. Thus every interior or boundary center has an affine approximation satisfying (2.9), with constant $C(L + M)$, for every $0 < \rho < \rho_*$. When $d(x) \geq r$, use $P_{x,\rho}(y) = u(x) + Du(x) \cdot (y - x)$ from (5.2). Lemma 2.5, with radius threshold ρ_*, now gives a continuous gradient field on the closed patch, equal to Du in the interior and Hölder continuous with exponent β for nearby pairs. At a boundary point z, (1.7) is a first-order expansion because $\lambda > 0$; uniqueness in that lemma identifies the boundary field with the prescribed p_z.

The ambient Lipschitz bound gives $|Du| \leq L$ in the interior, and continuity gives the same bound on the extended field. The larger-separation conclusion of Lemma 2.5 therefore supplies the full gradient seminorm on every fixed smaller closed patch. Together with the first-order expansions, this proves (1.8). □

Remark 5.1. A boundary expansion at a single point and a uniform expansion over a patch have different consequences. Under the $C^{1,\gamma}$ geometry and trace hypotheses of Section 4, (4.2) gives an anchored gradient estimate at a single point from a known rate for E_z. Theorem 1.4 uses uniform boundary expansions and produces a full gradient seminorm, including interior pairs with separation much smaller than their boundary distance.

6. Locally constant boundary traces

6.1. Odd reflection on a flat patch.

Lemma 6.1 (Odd reflection)**.** *Let u be infinity harmonic in $B_1^+ = B_1 \cap \{x_n > 0\}$, continuous on its closure, and zero on the flat face. Its odd extension*

$$U(x', t) = \begin{cases} u(x', t), & t \geq 0, \\ -u(x', -t), & t < 0 \end{cases}$$

is infinity harmonic in B_1.

Proof. Away from the reflecting plane, orthogonal invariance and sign invariance of the homogeneous equation apply. Suppose a C^2 function ϕ touches U from above at $(x_0', 0)$. Since $U = 0$ on the plane, $D'\phi(x_0', 0) = 0$. Oddness on the normal line and the touching inequality imply $\phi(x_0', t) + \phi(x_0', -t) - 2\phi(x_0', 0) \geq 0$. Consequently $\phi_{nn}(x_0', 0) \geq 0$, and

$$-\Delta_\infty \phi = -\phi_n^2 \phi_{nn} \leq 0.$$

A lower test gives the reversed signs. These are all the viscosity conditions on the plane. □

The flat conclusion of Theorem 1.5 follows immediately by subtracting the constant trace, applying Lemma 6.1, and using (H). Indeed, the reflected function has $\operatorname{osc}_{B_1} U = 2\|u - c\|_{L^\infty(B_1^+)}$. The full-ball estimate controls its gradient across the flat face, and restriction to the closed upper half-ball gives (1.10).

6.2. **Comparison with odd full-ball replacements.**

Lemma 6.2 (Replacement near a curved zero-trace patch)**.** *Assume* (H). *Suppose* $\Gamma \in C^{1,\gamma}$ *and* u *is infinity harmonic near* Γ*, continuous up to it with zero trace. On a fixed smaller neighborhood, there is a local ambient Lipschitz constant* L *and, for each boundary center* z *and every sufficiently small* s*, an infinity-harmonic function* $H_{z,s}$ *in the full ball* $B_s(z)$ *such that*

$$\|u - H_{z,s}\|_{L^\infty(\Omega \cap B_s(z))} \le CLs^{1+\gamma}, \tag{6.1}$$

$$\|DH_{z,s}\|_{L^\infty(B_{s/2}(z))} + s^a [DH_{z,s}]_{C^{0,a}(B_{s/2}(z))} \le CL. \tag{6.2}$$

The constants depend on n, γ, a, C_H *and the fixed chart bounds.*

Proof. Lemma 2.2 gives the bound for u on a larger controlled chart neighborhood. Choose a uniform $s_0 > 0$ so that $B_{2s}(z)$ and all graph points used below stay in that neighborhood whenever z lies in the smaller patch and $0 < s < s_0$. Extend u by zero to the other side of the boundary in a larger local ball, obtaining U_0. This extension is ambient Lipschitz: if $x \in \Omega$ and $y \notin \Omega$ are in this ball, their segment has a boundary crossing w, so $|u(x)| = |u(x) - u(w)| \le L|x - w| \le L|x - y|$. For two interior points the local ambient Lipschitz estimate already applies, and for two exterior points the difference vanishes. We choose the local ball convex and within the controlled neighborhood so that these crossings remain on the zero-trace patch.

Use tangent coordinates at $z = 0$, writing $\Omega = \{t > \psi(x')\}$, with $|\psi(x')| \le C|x'|^{1+\gamma}$. On the upper half of ∂B_s prescribe

$$\Phi(x', t) = U_0(x', t) - U_0(x', 0), \qquad t \ge 0,$$

and extend Φ oddly to the lower half. It is continuous and vanishes on the equator. Let H_s be its unique infinity-harmonic extension to B_s. Uniqueness gives oddness of H_s: the function $-H_s(x', -t)$ solves the same full-ball problem with the same spherical data. In particular, $H_s = 0$ on the flat disk. Since $|\Phi(x', t)| \le L|t|$, comparison with $\pm Lt$ in the upper half-ball, and oddness, give

$$|H_s(x', t)| \le L|t|. \tag{6.3}$$

On the true boundary, (6.3) bounds $|u - H_s|$ by $CLs^{1+\gamma}$. On the upper spherical part lying in Ω, the difference is $U_0(x', 0)$. It has size at most $L|\psi(x')|$ because U_0 vanishes at $(x', \psi(x'))$. On any lower spherical part lying in Ω, one has $\psi(x') < t < 0$, so $|t| \le Cs^{1+\gamma}$. Both $|H_s|$ and $|u|$, the latter compared with its zero value on the graph, are then at most $CLs^{1+\gamma}$. The graph points used in these estimates may lie just outside B_s but are in the larger Lipschitz neighborhood. Thus $H_s \pm CLs^{1+\gamma}$ order u on the entire boundary of $\Omega \cap B_s$, including the intersections of the graph and sphere by continuity. These functions remain infinity harmonic, so comparison proves (6.1).

Finally, (6.3) gives $\operatorname{osc}_{B_s} H_s \le 2Ls$. Apply (H) to obtain (6.2). □

Proof of the curved-boundary assertion of Theorem 1.5. Subtract the constant trace. Normalize a fixed neighborhood and let L be the Lipschitz constant from Lemma 2.2. Use the scale choice from (5.1) with $\lambda = \gamma$:

$$r = \rho^{(1+a)/(1+a+\gamma)}.$$

Choose a uniform $\rho_* > 0$ so that $0 < \rho < \rho_*$ implies $r \ge 8\rho$ and $8r < s_0$, where s_0 is the replacement threshold. As in Section 5, leave a fixed chart margin so that all centers, nearest

boundary points, and replacement balls lie in the controlled larger neighborhood. The deep-interior estimate (5.2), whose proof uses only (H) and the Lipschitz bound, covers every center with $d(x) \geq r$.

If $d(x) < r$, take a nearest boundary point z; for a boundary center take $z = x$. Set $H = H_{z,8r}$. Every point in the target clipped ball and its center lie in $B_{2r}(z)$, once $r \geq 8\rho$. Since this ball is convex, each segment from x to a point y of the target set lies in $B_{2r}(z) \subset B_{4r}(z)$, where (6.2) controls DH. Define

$$P_{x,\rho}(y) = u(x) + DH(x) \cdot (y - x).$$

The exact identity

$$\begin{aligned} u(y) - P_{x,\rho}(y) = (u - H)(y) - (u - H)(x) \\ + H(y) - H(x) - DH(x) \cdot (y - x) \end{aligned}$$

and Lemma 6.2 give

$$\begin{aligned} \sup_{\overline{\Omega} \cap B_\rho(x)} |u - P_{x,\rho}| &\leq 2CL(8r)^{1+\gamma} + CL\rho^{1+a}(8r)^{-a} \\ &\leq CL(r^{1+\gamma} + \rho^{1+a} r^{-a}). \end{aligned}$$

The replacement error extends to the closed domain by continuity, and H is defined on the full ball, so the argument includes boundary centers. The balancing calculation in Section 5, with $\lambda = \gamma$, makes both terms of order $\rho^{1+\beta}$, where $\beta = a\gamma/(1 + a + \gamma)$. Thus (2.9) holds uniformly at every center for all $0 < \rho < \rho_*$. Apply Lemma 2.5, including its larger-separation conclusion using the bound L for the extended gradient as in Section 5. This proves the full closure estimate.

Since the trace is zero, Lemma 2.2 gives $L \leq C \|u\|_{L^\infty}$ in normalized coordinates, with geometry fixed. All the preceding error and gradient estimates are linear in L. Returning to the original function and physical scale R, the chain rule therefore gives

$$R \|Du\|_{L^\infty(\text{smaller patch})} + R^{1+\beta}[Du]_{C^{0,\beta}(\text{smaller patch})} \leq C \|u - c\|_{L^\infty(\text{larger patch})}.$$

□

Remark 6.3 (Local limitation on the exponent). The example $u(x, y) = |x|^{4/3} - |y|^{4/3}$ is infinity harmonic and vanishes on the line $y = x$. In the half-plane $y > x$, $Du(0, t) = (0, -\frac{4}{3}t^{1/3})$ for $t > 0$, while $Du(0) = 0$. Thus a universal local boundary exponent greater than $1/3$ cannot hold even with a flat patch and locally constant trace. To verify the viscosity equation, the two classical contributions away from the axes are $64/81$ and $-64/81$, so they cancel. At $(0, y_0)$ with $y_0 \neq 0$, the positive $|x|^{4/3}$ term excludes an upper C^2 test. A lower test has $\phi_x = 0$ and, by restriction to the y direction,

$$\phi_{yy} \leq -\tfrac{4}{9}|y_0|^{-2/3}, \qquad -\Delta_\infty \phi = -\phi_y^2 \phi_{yy} \geq 0.$$

At $(x_0, 0)$ with $x_0 \neq 0$, the negative $|y|^{4/3}$ term excludes a lower C^2 test. An upper test has $\phi_y = 0$ and

$$\phi_{xx} \geq \tfrac{4}{9}|x_0|^{-2/3}, \qquad -\Delta_\infty \phi = -\phi_x^2 \phi_{xx} \leq 0.$$

Finally u is differentiable at the origin with gradient zero, so every touching test has zero gradient and satisfies the homogeneous viscosity inequality automatically. This verifies the example on all its singular sets.

7. Nonconstant data with transverse calibration

Lemma 7.1 (Global Euclidean Lipschitz bound). *Let Ω be bounded and $u \in C(\overline{\Omega})$ solve (1.1). If $L = \mathrm{Lip}(g, \partial\Omega) < \infty$, then $\mathrm{Lip}(u, \overline{\Omega}) \leq L$.*

Proof. For every $z \in \partial\Omega$, compare with $g(z) \pm L|x-z|$ to get $|u(x)-g(z)| \le L|x-z|$ throughout Ω. Fix $x \in \Omega$. The cones $u(x) \pm L|y-x|$ therefore order g on $\partial\Omega$. They are continuous on the closure of the bounded punctured domain $\Omega \setminus \{x\}$ and infinity harmonic there. At its added boundary point x they both equal $u(x)$. The bounded-domain comparison from Section 2 therefore gives directly $|u(y) - u(x)| \le L|y - x|$. Continuity gives boundary pairs. Convexity is not needed for this lemma. □

Proof of Theorem 1.6. Fix a boundary point z. Replace u, g temporarily by $\sigma_z u, \sigma_z g$ so that the calibration has positive sign. Lemma 7.1 applies with the same constant L.

The boundary function

$$q \longmapsto g(b_z) - g(q) - L|b_z - q|$$

is nonpositive and vanishes at z. Its tangential derivative vanishes there. Since $b_z \neq z$, the tangential derivative of $|b_z - q|$ at $q = z$ is $-P_{T_z\partial\Omega} e_z$, giving

$$D_T g(z) = L P_{T_z\partial\Omega} e_z. \tag{7.1}$$

No boundary derivative of u is used in this step.

The Lipschitz inequality based at b_z gives $u(x) \ge g(b_z) - L|x - b_z|$. Since $|b_z - z| \ge \ell$, the function $|x - b_z|$ has Hessian norm at most C/ℓ along the segment from z to x when $|x - z| \le \ell/4$. Taylor expansion therefore gives

$$u(x) \ge g(z) + Le_z \cdot (x - z) - C(L/\ell)|x - z|^2. \tag{7.2}$$

For the upper bound, use tangent coordinates at $z = 0$ and write the graph as $x_n = \psi(x')$. Uniform $C^{1,\gamma}$ bounds allow the chart to be made small so that $|D\psi| \le \tau/2$. For x in the interior near z consider

$$F(t) = x_n - te_{z,n} - \psi(x' - te'_z).$$

One has $F(0) > 0$ and $F'(t) = -e_{z,n} + D\psi(x' - te'_z) \cdot e'_z \le -\tau/2$ while the line remains in the chart. To ensure that it stays there until it meets the boundary, put $t_* = 2F(0)/\tau$. Since $\psi(0) = 0$ and $|D\psi| \le \tau/2$, one has $F(0) \le C|x - z|$. A uniform sufficiently small bound on $|x-z|$ therefore keeps the whole segment $x - te_z$, $0 \le t \le t_*$, inside the chart: its distance to z is at most $|x - z| + t_* \le C_\tau|x - z|$. Integrating $F' \le -\tau/2$ gives $F(t_*) \le 0$. The intermediate value theorem supplies a zero $0 < t \le t_* \le C_\tau|x - z|$. Thus $q = x - te_z$ is a boundary point with $|q - z| \le C_\tau|x - z|$. The Lipschitz estimate gives $u(x) \le g(q) + Lt$.

Write $G(\xi) = g(\xi, \psi(\xi))$ in these tangent coordinates. Equation (7.1) gives $DG(0) = Le'_z$. Consequently, for $q = (q', \psi(q'))$,

$$\begin{aligned} g(q) - g(z) - Le_z \cdot (q - z) = G(q') - G(0) - Le'_z \cdot q' \\ - Le_{z,n}\psi(q'). \end{aligned}$$

The $C^{1,\gamma}$ Taylor bounds for G and ψ imply

$$|g(q) - g(z) - Le_z \cdot (q - z)| \le C|q - z|^{1+\gamma}.$$

Since $x - q = te_z$, we conclude that

$$u(x) \le g(z) + Le_z \cdot (x - z) + C|x - z|^{1+\gamma}. \tag{7.3}$$

The quadratic error in (7.2) is bounded by the same order at small radii because $\gamma \le 1$. Undoing the temporary sign change gives (1.13), extending to boundary points x by continuity. All radii and constants are uniform on smaller patches: the argument uses only their common chart and trace bounds, L, ℓ^{-1}, and τ^{-1}. Theorem 1.4, with $\lambda = \gamma$, proves the stated full Hölder estimate. □

Remark 7.2. The proof of the boundary expansion uses only local transversality, the global Lipschitz estimate, and the calibrating boundary pair. Convexity in the theorem makes the chord interpretation immediate. No regular selection $z \mapsto b_z$ or family of gradient-flow curves is required.

7.1. **Examples and the role of the data condition.** For a smooth nonaffine example, let $\Omega = B_1$ and $g(z) = |z - Ae_1|$, $A > 2$. The solution is the cone $u(x) = |x - Ae_1|$, whose vertex lies outside the ball. The cone and uniqueness facts in Section 2 justify this identification. The boundary Lipschitz constant is 1: it is at most 1 by the triangle inequality, while the pair $e_1, -e_1$ gives $g(-e_1) - g(e_1) = 2 = |-e_1 - e_1|$. The trace is nonaffine because its sums on the antipodal pairs $\pm e_1$ and $\pm e_2$ are $2A$ and $2\sqrt{A^2+1}$, whereas an affine trace would have the same sum on every such pair. On the visible cap $z_1 > 1/A$ the direction

$$e_z = \frac{z - Ae_1}{|z - Ae_1|}$$

satisfies

$$e_z \cdot \nu(z) = \frac{Az_1 - 1}{\sqrt{A^2 + 1 - 2Az_1}} > 0.$$

The second intersection is $b_z = z + t_z e_z$, where solving $|z + t_z e_z| = 1$ gives

$$t_z = -2z \cdot e_z = \frac{2(Az_1 - 1)}{|z - Ae_1|} = 2e_z \cdot \nu(z) > 0.$$

Since $b_z - Ae_1 = (|z - Ae_1| + t_z)e_z$, one has $g(b_z) - g(z) = t_z = |b_z - z|$. On every compact subcap the displayed dot product has a positive minimum, giving both uniform transversality and a uniform positive chord length. Thus the hypotheses hold on every smaller open cap whose closure stays in the visible cap.

The domain condition alone does not supply calibration. On the unit circle take $g(\cos\vartheta, \sin\vartheta) = \cos^2\vartheta$. For distinct points,

$$\frac{|g(\vartheta) - g(\varphi)|}{|(\cos\vartheta, \sin\vartheta) - (\cos\varphi, \sin\varphi)|} = |\sin(\vartheta + \varphi)| \left|\cos\frac{\vartheta - \varphi}{2}\right| < 1.$$

Distinct physical points have $\vartheta - \varphi \notin 2\pi\mathbb{Z}$, so the cosine factor is strictly less than 1 in absolute value. Taking $\vartheta = \pi/4 + h$ and $\varphi = \pi/4 - h$, with $0 < h < \pi/2$, gives quotient $\cos h \to 1$ as $h \downarrow 0$. Thus the Lipschitz constant is 1, but no distinct pair attains it. The analytic strictly convex disk with analytic data need not satisfy (1.11). This example concerns the calibration condition and makes no claim of irregularity for its solution.

8. The remaining unrestricted Hölder problem

For arbitrary $C^{1,\gamma}$ data on a $C^{1,\gamma}$ patch, Theorem 1.2 and (4.2) prove boundary continuity of the gradient. To conclude a positive spatial Hölder exponent by the present route, it would be enough to prove, uniformly on each fixed smaller boundary patch Γ', the existence of common $C > 0$, $0 < r_0 \le 1$, and $\lambda > 0$ such that

$$E_z(r) \le Cr^\lambda \quad (z \in \Gamma',\ 0 < r < r_0), \tag{8.1}$$

where p_z and E_z are those in (4.1). If this estimate holds, put $\lambda_0 = \min\{\lambda, 1\}$. Multiplying (8.1) by r gives

$$\sup_{\overline{\Omega} \cap B_r(z)} |u(x) - g(z) - p_z \cdot (x - z)| \le Cr^{1+\lambda} \le Cr^{1+\lambda_0}.$$

This is (1.7), while Lemma 2.2 supplies the required local Lipschitz bound. Theorem 1.4 would then give $\beta = a\lambda_0/(1 + a + \lambda_0)$.

An affine residual cannot be inserted directly into the original interior estimate. For $\ell(x) = c + p \cdot x$, $0 < r < 1$, and a proposed exponent $\beta > 0$, set

$$w_r(x) = \frac{u(rx) - \ell(rx)}{r^{1+\beta}}.$$

A C^2 test ϕ touching w_r corresponds to the test $\Phi(y) = \ell(y) + r^{1+\beta}\phi(y/r)$ for u, with the same type of contact. At $y = rx$ its derivatives are

$$D\Phi(rx) = p + r^\beta D\phi(x), \qquad D^2\Phi(rx) = r^{\beta-1} D^2\phi(x).$$

The factor $r^{3\beta-1}$ in the resulting infinity Laplacian is positive. Thus the actual equation, in the viscosity sense wherever $rx \in \Omega$, is

$$\left\langle D^2 w_r (Dw_r + r^{-\beta} p),\, Dw_r + r^{-\beta} p \right\rangle = 0. \tag{8.2}$$

The restored function

$$U_r(x) = w_r(x) + r^{-\beta} p \cdot x = \frac{u(rx) - c}{r^{1+\beta}}$$

is infinity harmonic. On any interior ball $B_R(x_0) \Subset r^{-1}\Omega$, (H) gives

$$\begin{aligned} [Dw_r]_{C^{0,a}(B_{R/2}(x_0))} &= [DU_r]_{C^{0,a}(B_{R/2}(x_0))} \\ &\le C_H R^{-1-a} \operatorname{osc}_{B_R(x_0)} U_r \\ &\le C_H R^{-1-a} \big(\operatorname{osc}_{B_R(x_0)} w_r + 2Rr^{-\beta}|p| \big). \end{aligned}$$

This direct application retains a term depending on the affine slope. Consequently (H) does not by itself justify an excess-decay iteration uniform in the shift $r^{-\beta}p$.

There is a similar limitation on reflection with an affine trace. For instance, in the upper half-disk $B_1^+ \subset \mathbb{R}^2$ let $u(x,y) = \sqrt{(A-x)^2 + y^2}$, $A > 2$, so that $u(x,0) = A - x$. This is an infinity-harmonic cone by Section 2, since its vertex $(A,0)$ lies outside the closed disk. The reflected expression

$$V(x,y) = 2(A-x) - \sqrt{(A-x)^2 + y^2} \quad (y < 0)$$

has the same trace $V(x,0) = A - x$, since $A - x > 0$ on the half-disk. Put $s = A - x$, $\varrho = (s^2 + y^2)^{1/2}$, and $e = (-s, y)/\varrho$. Since $A > 2$, $\varrho > 0$ throughout the comparison neighborhood. Direct differentiation of the reflected function gives

$$DV = (-2, 0) - e, \qquad D^2 V = -\frac{I - e \otimes e}{\varrho}.$$

Consequently

$$\Delta_\infty V = -\frac{|DV|^2 - (DV \cdot e)^2}{\varrho} = -\frac{4y^2}{((A-x)^2 + y^2)^{3/2}},$$

which is nonzero for $y < 0$. Thus the constant-trace reflection proof of Section 6 does not extend by this affine correction. General traces are treated by the separate barrier argument in Section 3.

The general-data theorem proved in this manuscript is the C^1 conclusion of Theorem 1.2, relative to the accepted uniform interior estimate. The positive Hölder conclusions are Theorems 1.4, 1.5, and 1.6, with their stated hypotheses. Estimate (8.1) is not established here for unrestricted Dirichlet data.

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